DTCP Firmware Updates

Dmytro Stavskyi

01/21/2026

Objective

- Be able to communicate with the AFE chip using SPI (receive any valid data from the chip, e.g., register values)

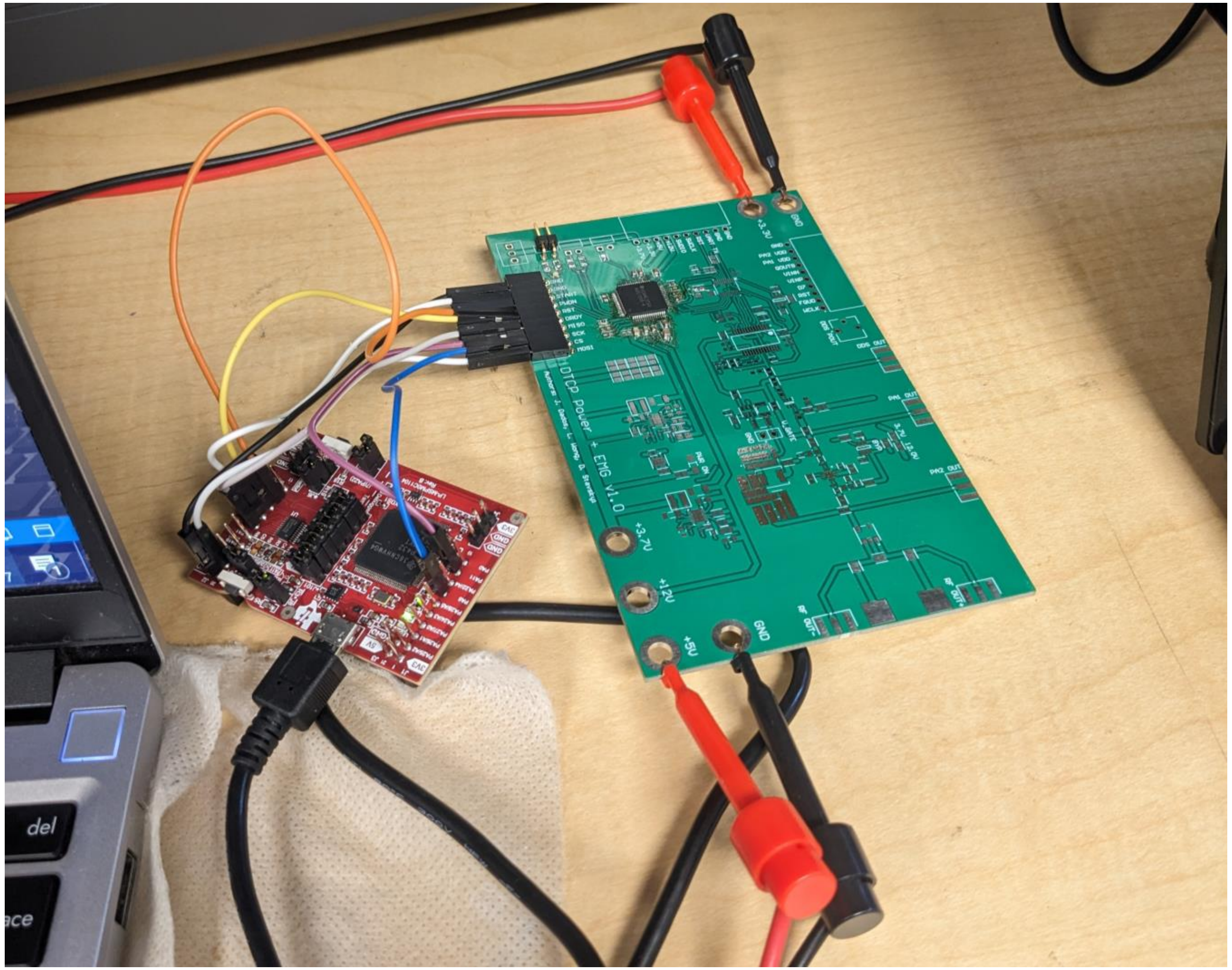
Methods

Assemble a separate board with only AFE in it for component isolation

Use the debug probs and the LaunchPad to program the chip

Measure the SCLK, CS and MOSI / MISO using oscilloscope

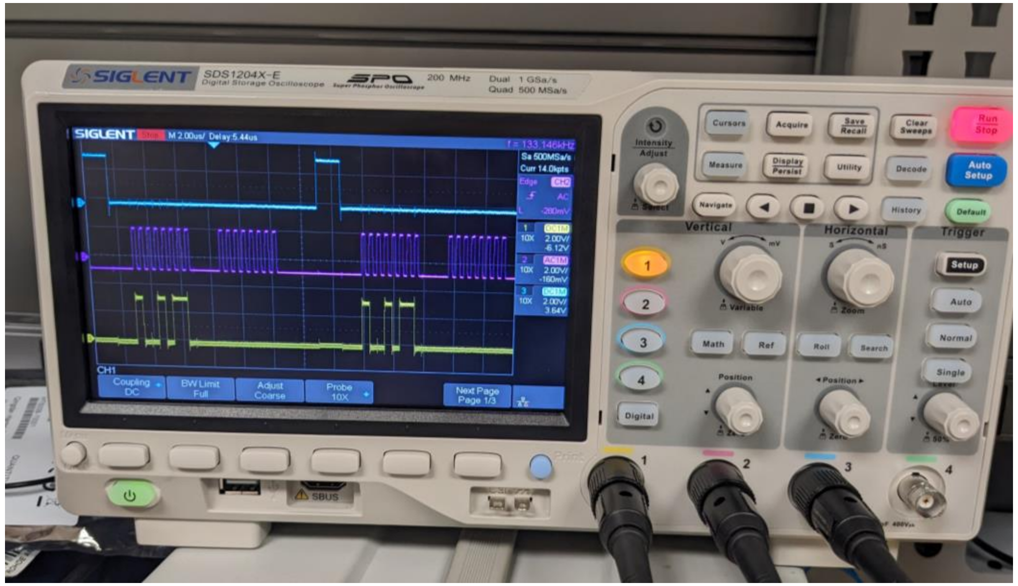
Measure the current consumption on the power supply

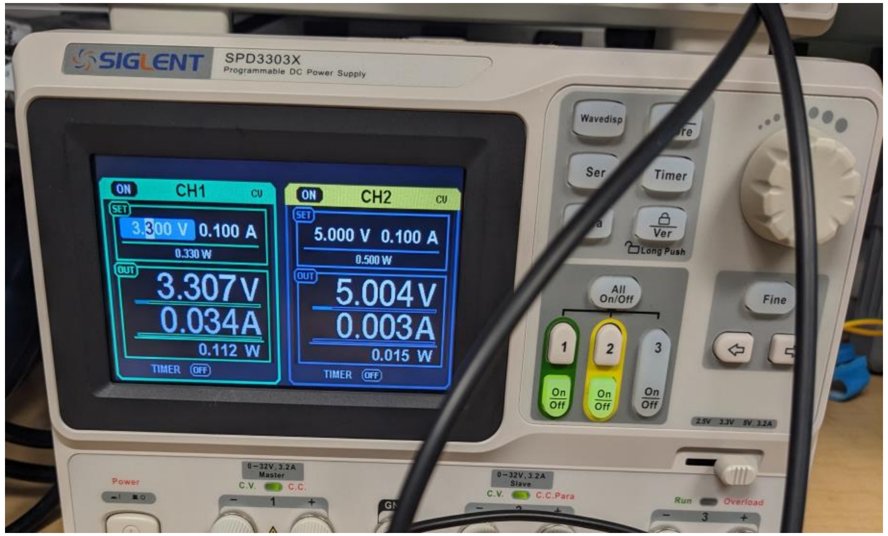


Results

At this point, we can write to registers and read from them and receive proper values

Due to incorrect configuration for one pin, the current consumption was 35 mA, but after the fix, it went back to 1~2 mA





Observation / Conclusion

- In RDATAC (Read Data Continuous) mode, you cannot read or write to registers
  - After issuing that mode using SPI, trying to read a register yields nothing, but DRDY (Data Ready) is never asserted
- Possible sources of errors: incorrect values for configuration registers

Next Steps

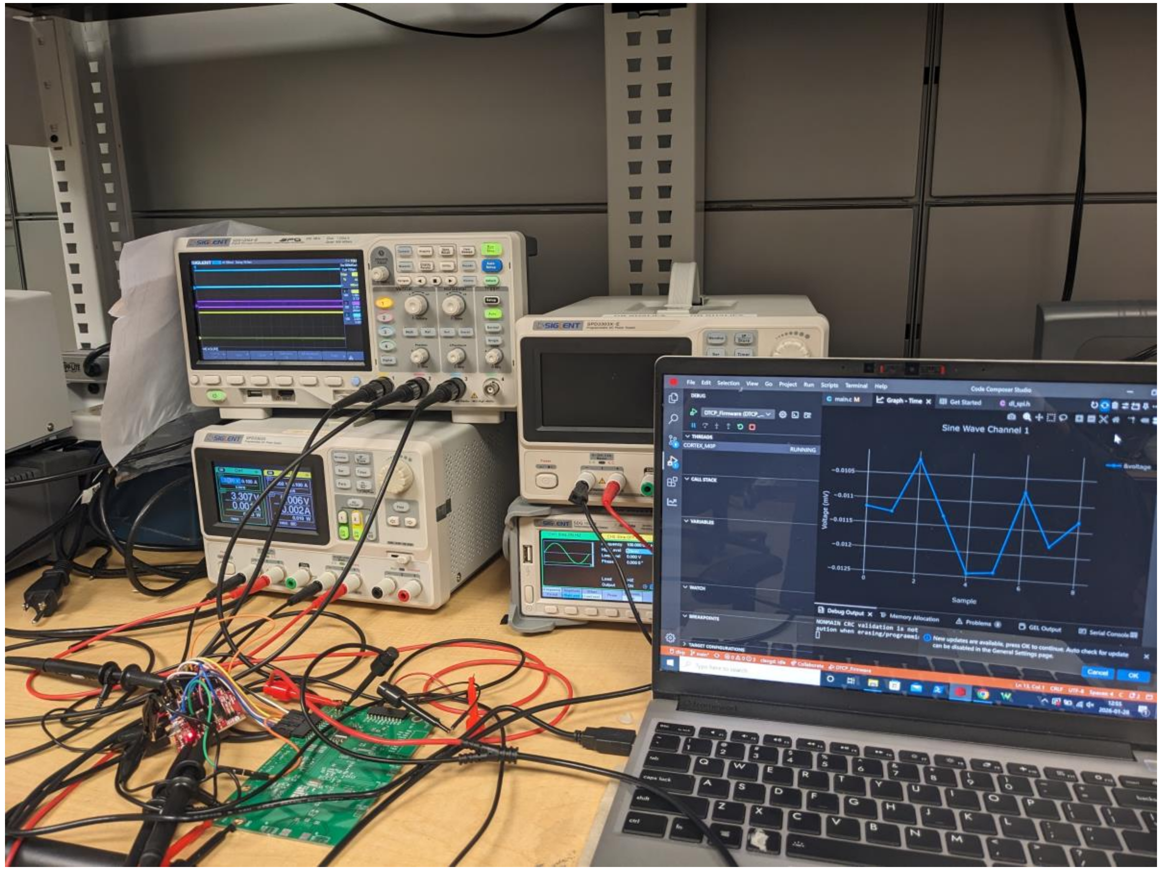
- Get conversion data from the analog channels

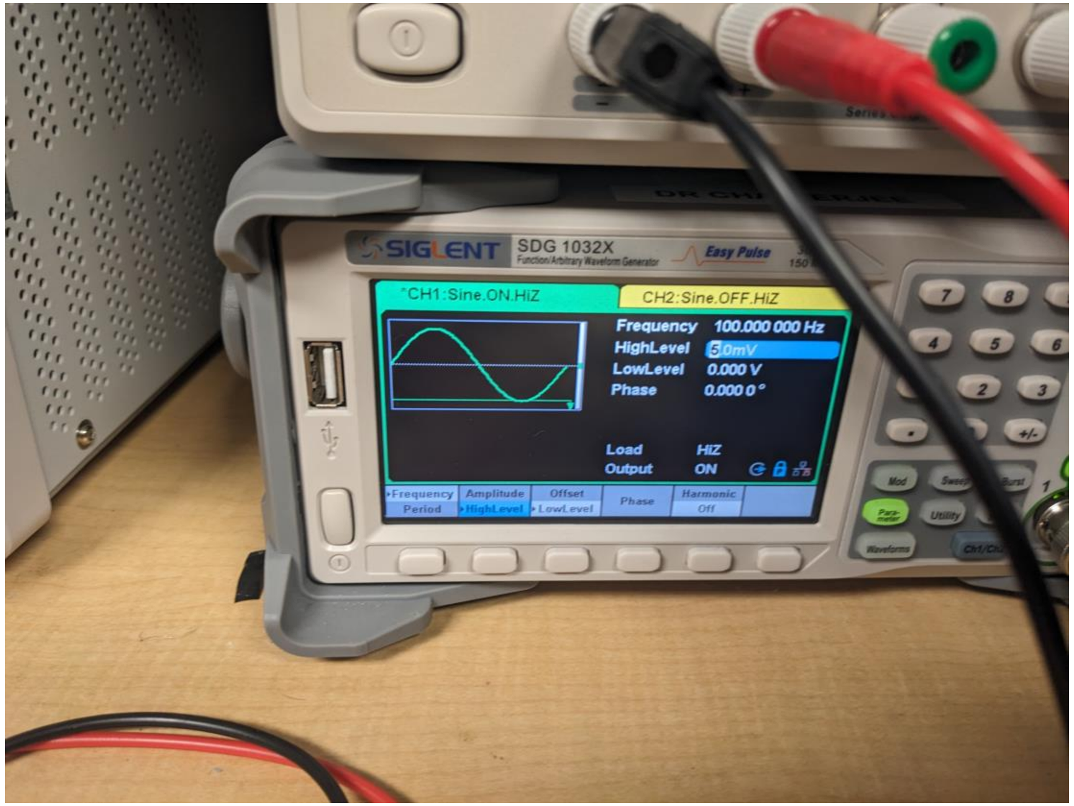
01/28/2026

Objective

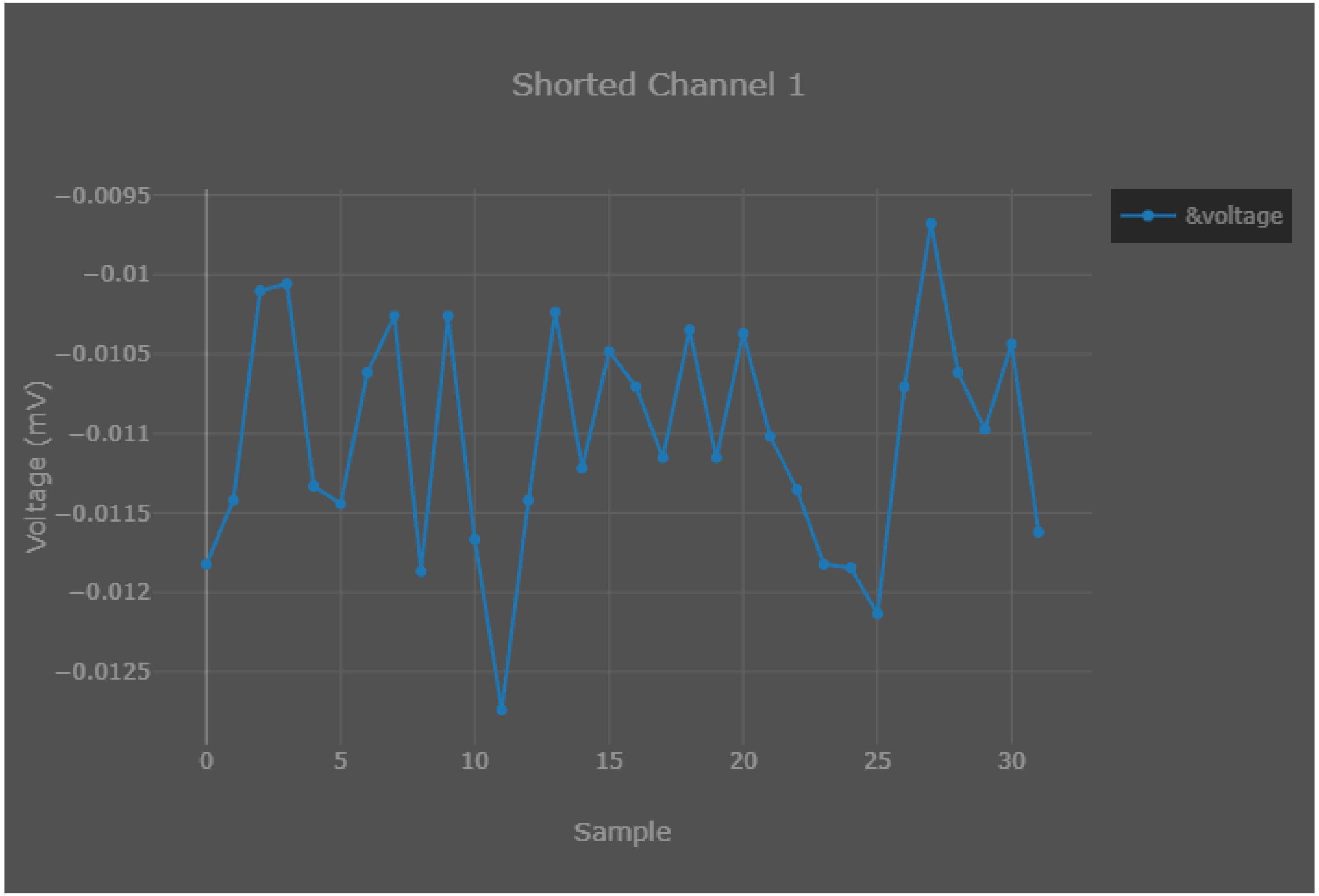
- Get conversion data from the analog channels

Methods

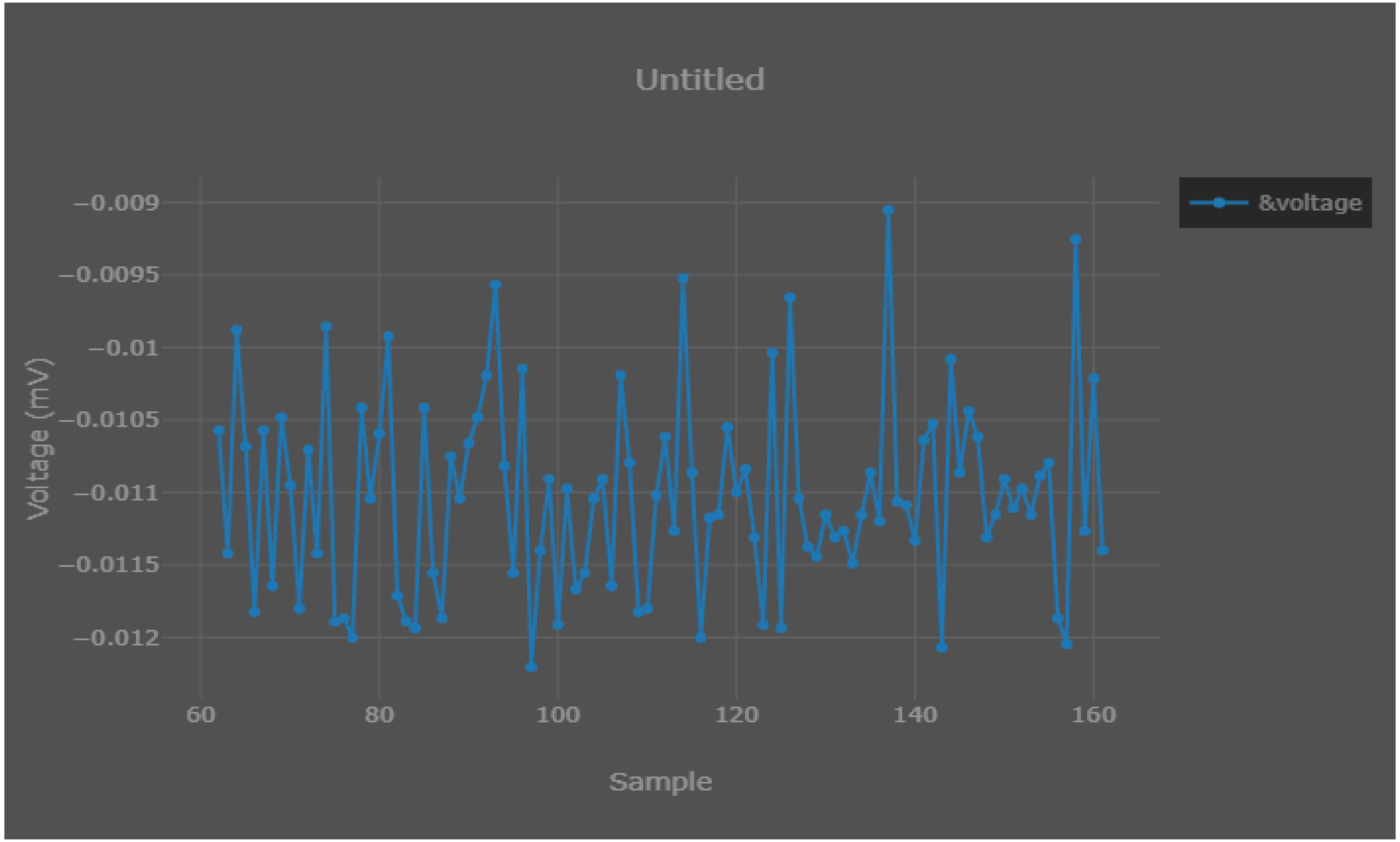

SIGLENT
Sine Wave Channel 1
Sample
Voltage (mV)


SIGLENT
SDG 1032X
Function/Arbitrary Waveform Generator
Easy Pulse
CH1:Sine.ON.HiZ
CH2:Sine.OFF.HiZ
Frequency 100.000 000 Hz
HighLevel 5.0mV
LowLevel 0.000 V
Phase 0.000 0 °
Load HiZ
Output ON
Frequency
Period
Amplitude
HighLevel
Offset
LowLevel
Phase
Harmonic
Off

Results (shorted inputs; nominal sampling rate of 1000)



Results (100 Hz 5 mV sine wave; nominal sampling rate of 1000)



Observation / Conclusion

- There were timing issues with the SPI, but adding some delays fixed the issues
- The first 24 bits are status bits, and they are what they should be, so we can assume that we are reading the data correctly

Next steps

- Improve firmware (interrupts, etc.)
- Test the conversion with different configuration values

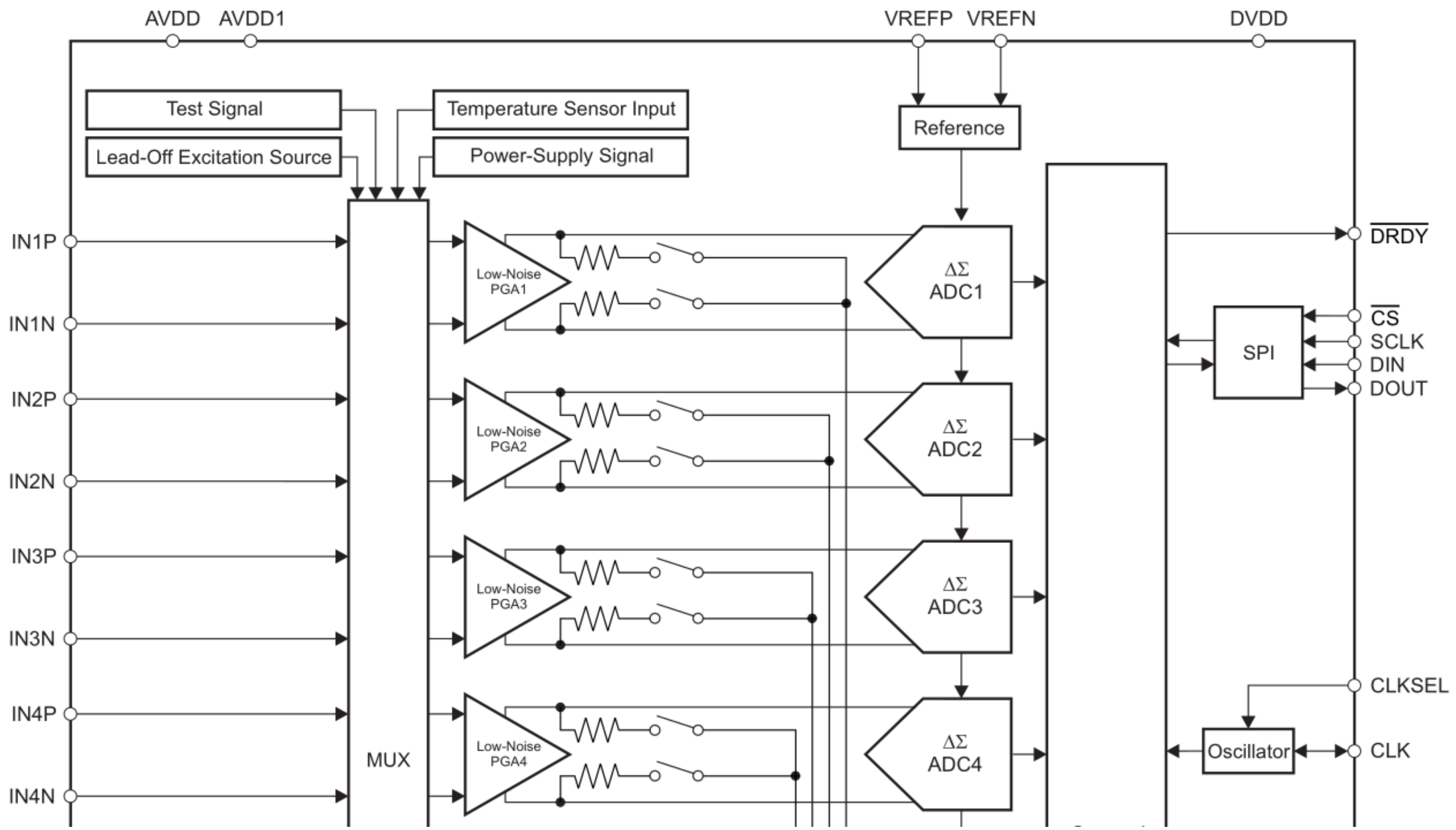
02/04/2026

Objective

- Determine if the chip can properly process any data
- Measure a wave provided by the function generator (e.g., 10 mV 100 Hz sine)

Methods

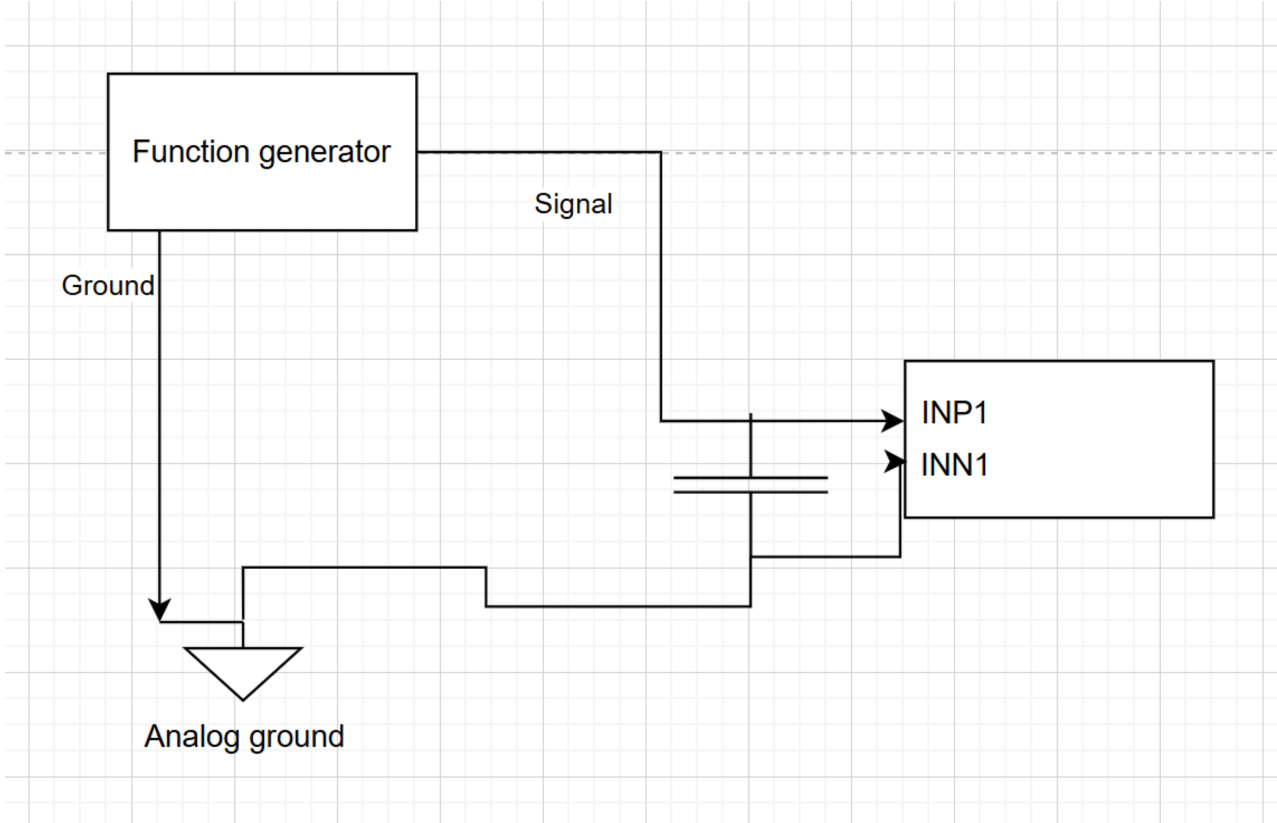
9.2 Functional Block Diagram



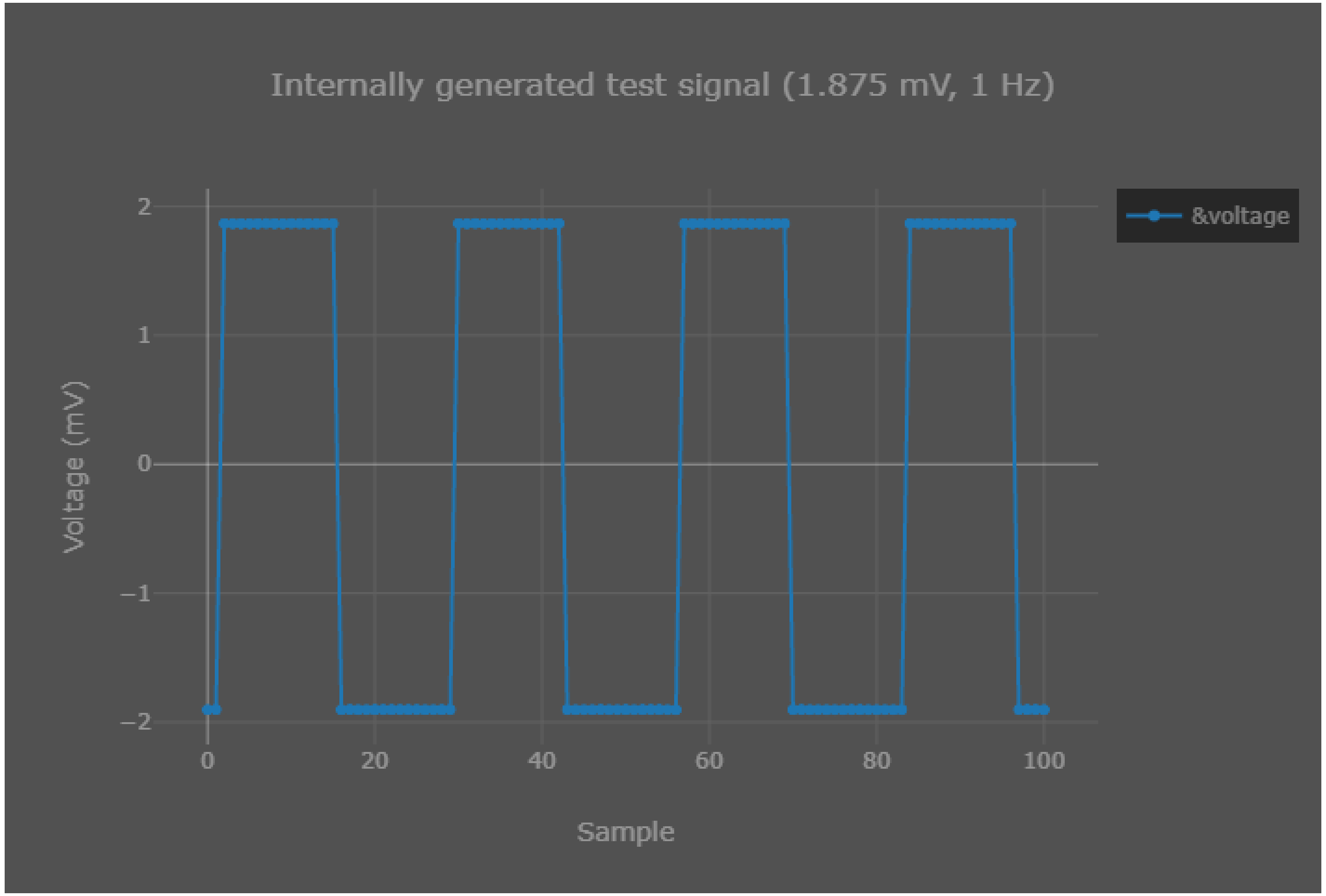
You can route the input to the channel from the normal external terminals to some other source like a test signal or a power-supply signal using a MUX

Methods


SIGLENT
SDS1204X-E
Digital Storage Oscilloscope
200 MHz
Dual 1 GSa/s
Quad 500 MSa/s
Intensity
Adjust
Navigate
Vertical
Math
Position
Digital


Function generator
Signal
Ground
INP1
INN1
Analog ground

Results (Test signal measurement)

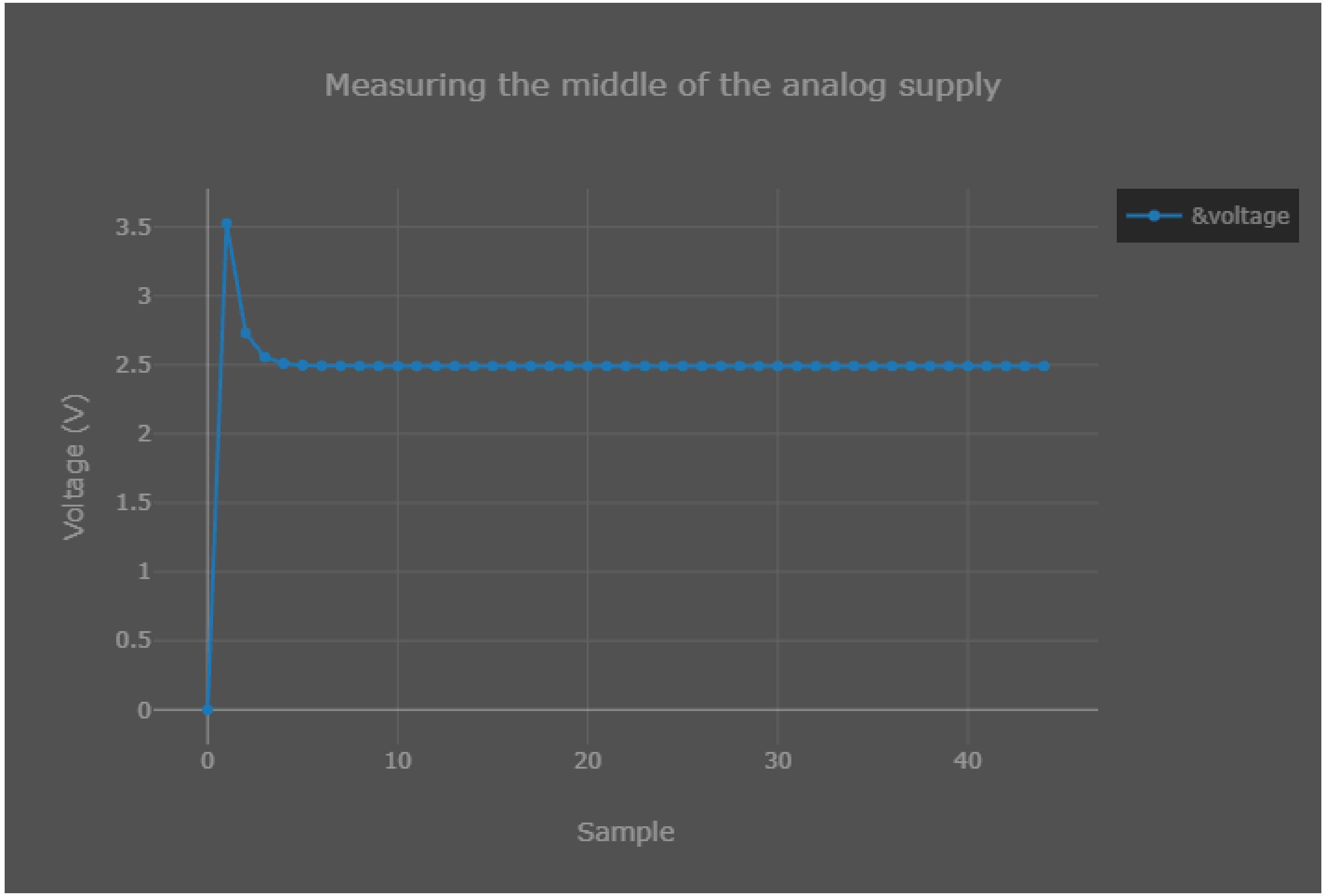


Top: 1.8487 mV

Bottom: -1.897537 mV

Measured Amplitude: 1.873 mV

Results (Supply measurement)



The input (based on the datasheet) is
(AVDD + AVSS) / 2
= (5 V + 0) / 2
= 2.5 V

Results (Sampling rate)

Frequency of the data ready signal (986.45 Hz) which is equivalent to the sampling rate (1 kHz)

Observation / Conclusion

- When using the normal input to the channel, the measurements are still essentially noise (even when I did a 1 Hz square wave)
- Since the test signal measurements are fine, the problem is probably to the left of the MUX, so the input terminals and/or function generator and/or grounding

Observation / Conclusion

9.3.1.2 Analog Input

The analog inputs to the device connect directly to an integrated low-noise, low-drift, high input impedance, programmable gain amplifier. The amplifier is located following the individual channel multiplexer.

The ADS1299-x analog inputs are fully differential. The differential input voltage ($V_{INxP} - V_{INxN}$) can span from $-V_{REF}$ / gain to V_{REF} / gain. See the *Data Format* section for an explanation of the correlation between the analog input and digital codes. There are two general methods of driving the ADS1299-x analog inputs: pseudo-differential or fully-differential, as shown in Figure 20, Figure 21, and Figure 22.

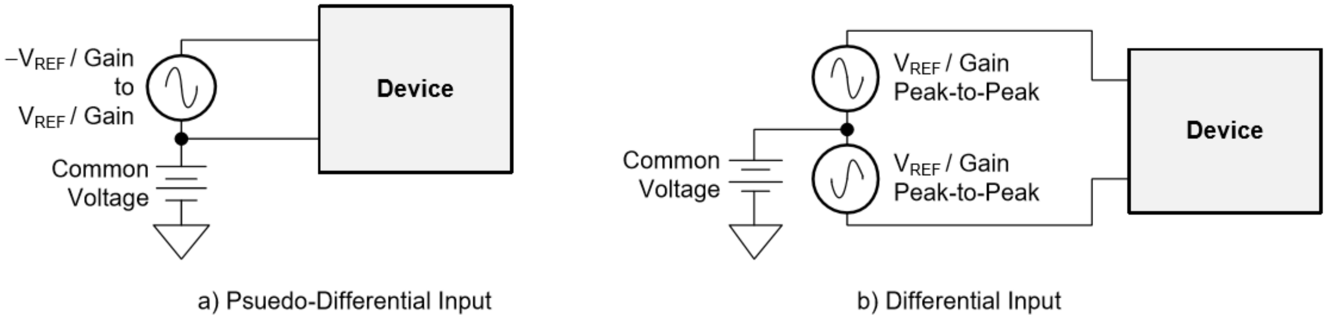


Figure 20. Methods of Driving the ADS1299-x: Pseudo-Differential or Fully Differential

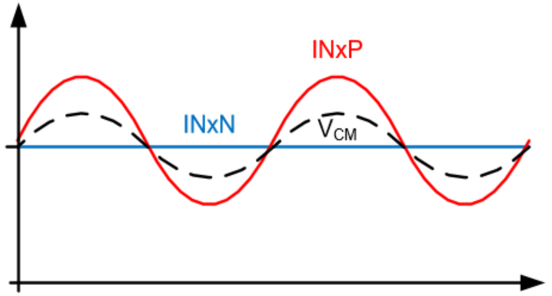


Figure 21. Pseudo-Differential Input Mode

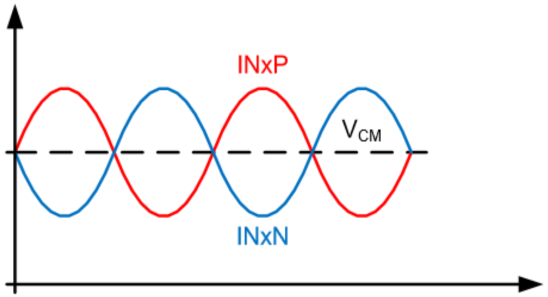


Figure 22. Fully-Differential Input Mode

The datasheet recommends the common voltage to be mid-supply (2.5 V in our case)

Vref / gain = 4.5 / 24 = 0.1875 V = 187.5 mV

Next steps

- Research the bias amplifier
- Test with a different channel

02/11/2026

Dmytro Stavskyi and Luis Wong

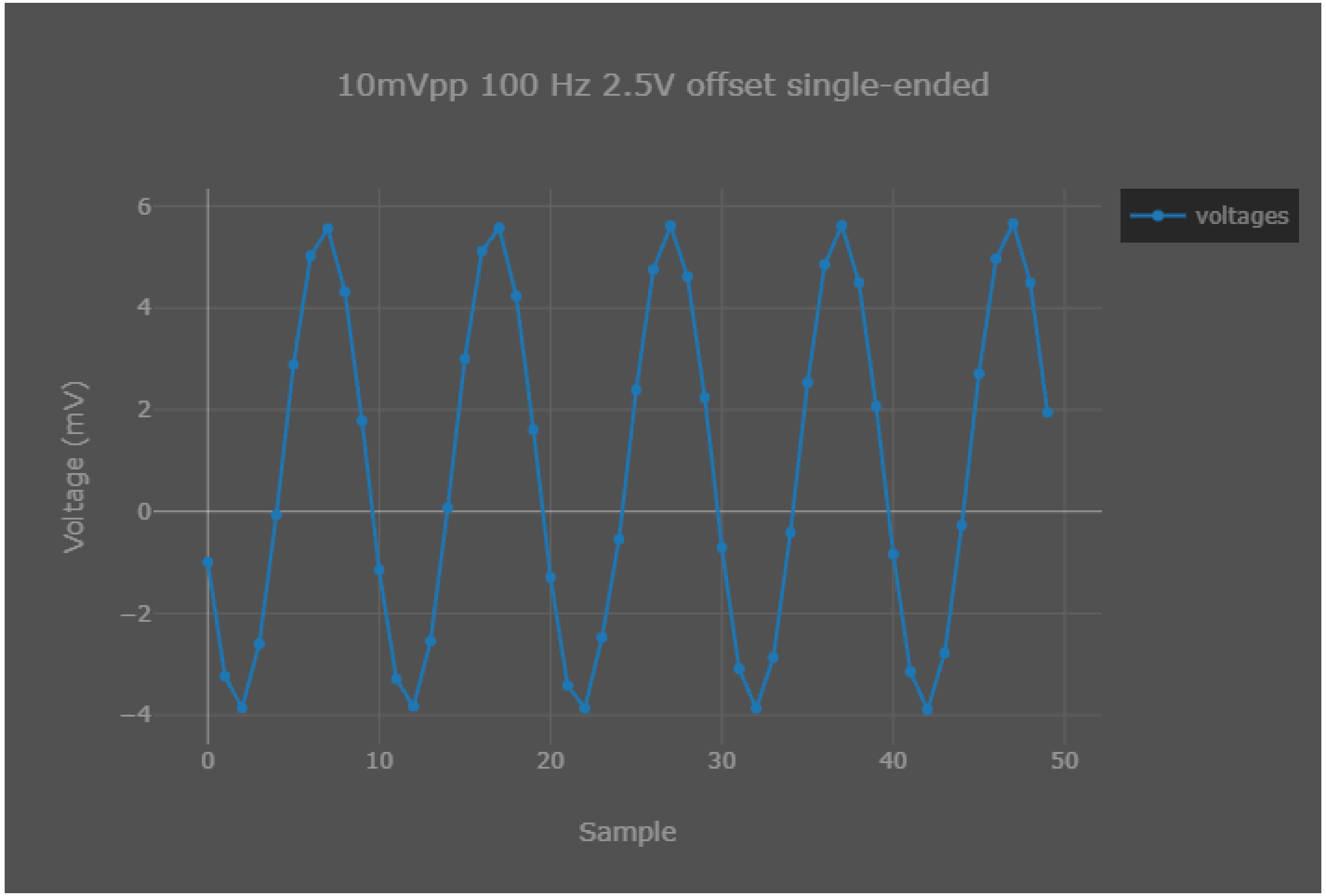
Objective

- Capture normal input data provided by the function generator

Methods

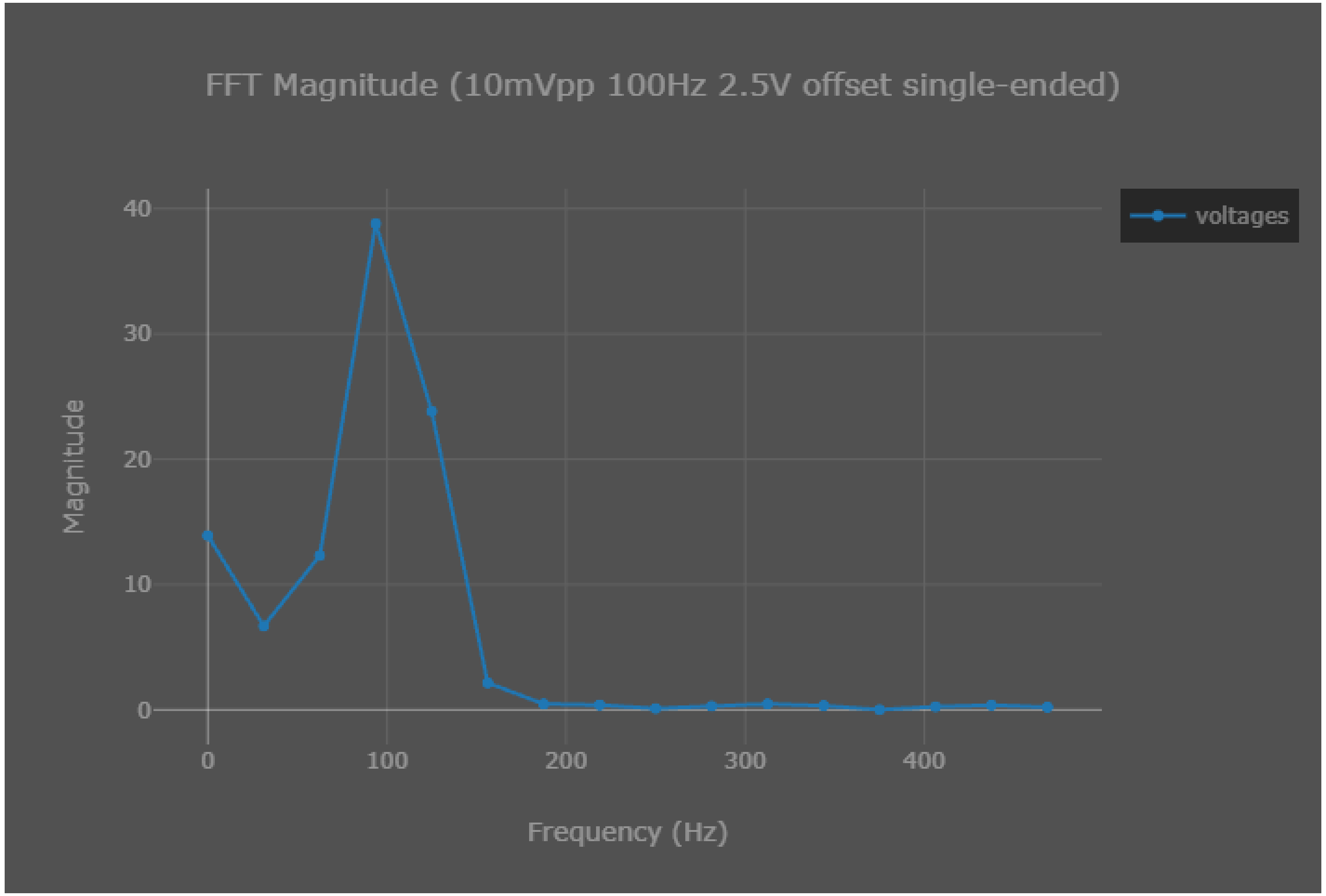
- Instead of continuously updating and plotting a variable, we save a number of points (e.g., 50) in an array and then plot those points
- Use FFT to calculate the frequencies of the captured signal
- For single-ended measurements, we applied the sine wave signal to the positive terminal (offset provided by the function generator) and either analog ground or some DC offset to the negative terminal

Results (5mV 100Hz 2.5 V offset single-ended)



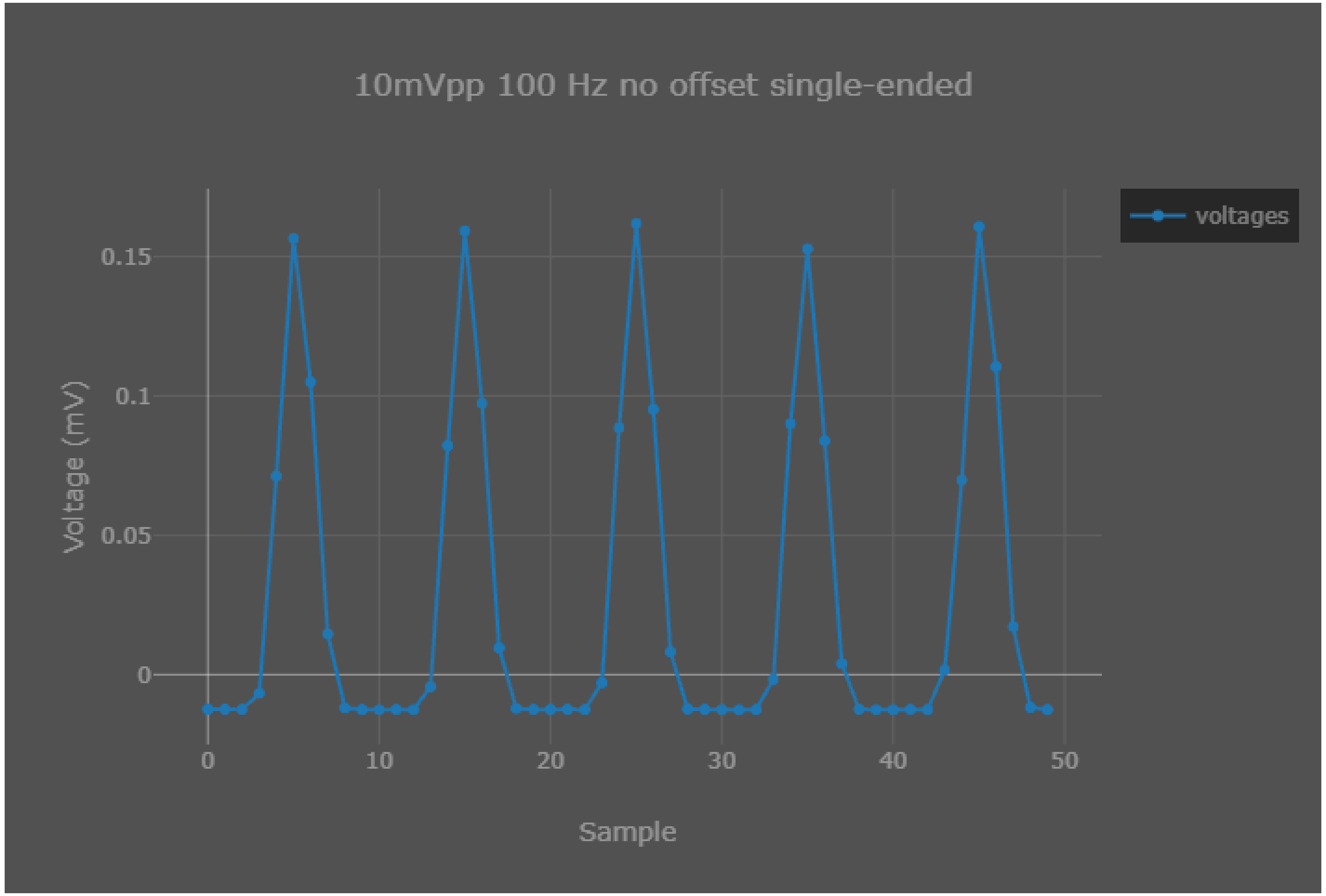
- Result was achieved by storing the data points in an array rather than a single variable.
- The graph is generated every time a breakpoint is reached, by default this is located at the location of the variable we are watching.
- According to the CCS documentation the graph runs through the debug probe and can only update every 0.1s under extremely ideal conditions.
- The [Sampling Rate] setting in CCS is arbitrary, and the true sampling rate is determined by the peripheral.

Results (5mV 100Hz 2.5 V offset single-ended FFT)



- Used an FFT to confirm that the frequency of the signal captured is 100Hz just like we generated.

Results (5mV 100 Hz no offset single-ended)



- No offset causes the device to essentially only capture noise.

The usable input common-mode range of the front-end depends on various parameters, including the maximum differential input signal, supply voltage, PGA gain, and the 200 mV for the amplifier headroom. This range is described in Equation 4:

$$AVDD - 0.2\ V - \left(\frac{Gain \times V_{MAX\_DIFF}}{2}\right) > CM > AVSS + 0.2\ V + \left(\frac{Gain \times V_{MAX\_DIFF}}{2}\right)$$

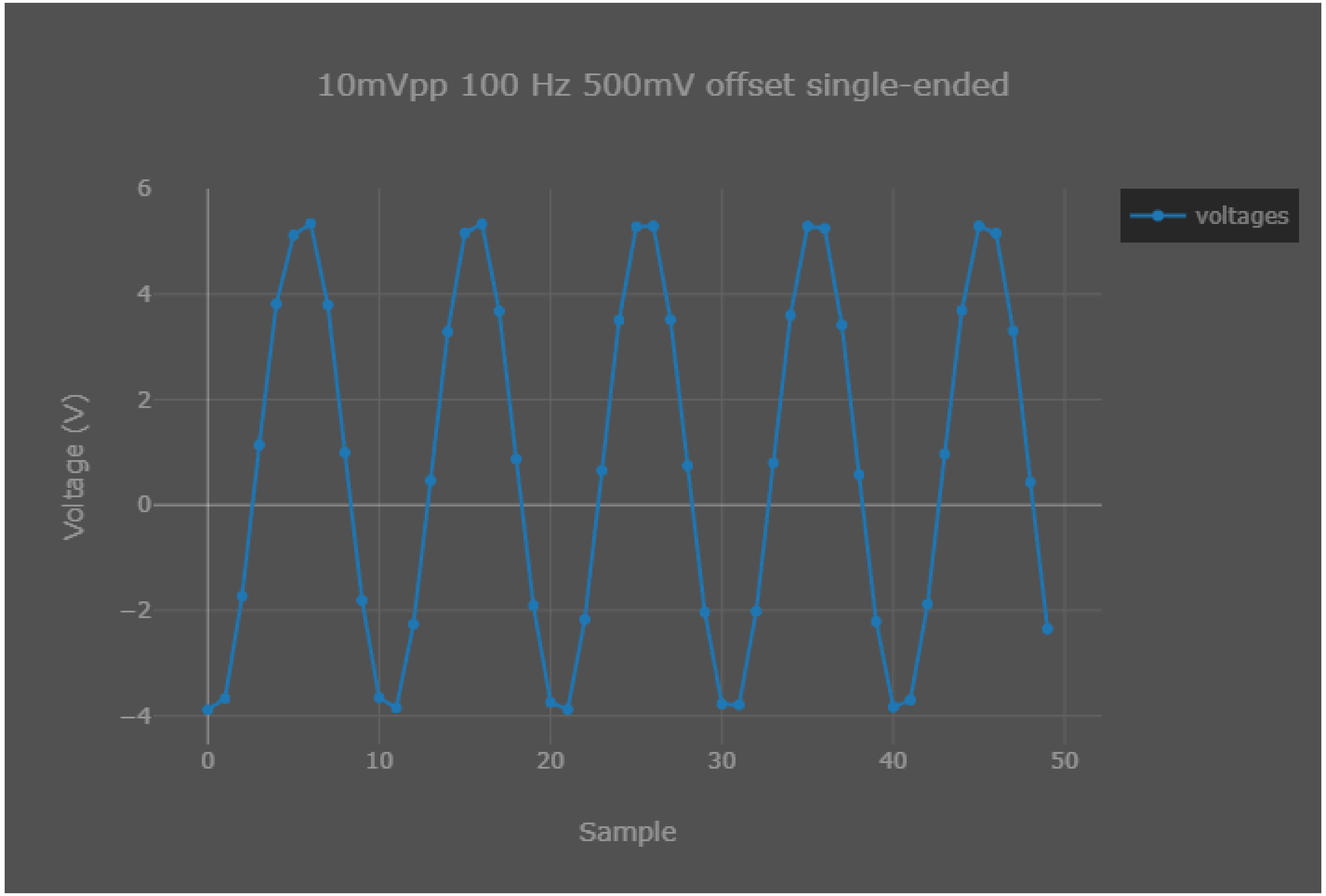
where:

V_{MAX_DIFF} = maximum differential signal at the PGA input

CM = common-mode range (4)

- AVDD = 5V
- Gain = 24
- AVSS = 0V
- V_{max_diff} = maximum differential signal at the PGA input = 10mV
- 0.32V < CM < 4.68V

Results (5mV 100 Hz 500mV offset single-ended)



Observation / Conclusion

- We need to offset the signal internally which can be achieved through the bias circuit.
- We did not wire the bias circuit when creating this PCB, so it may not work properly

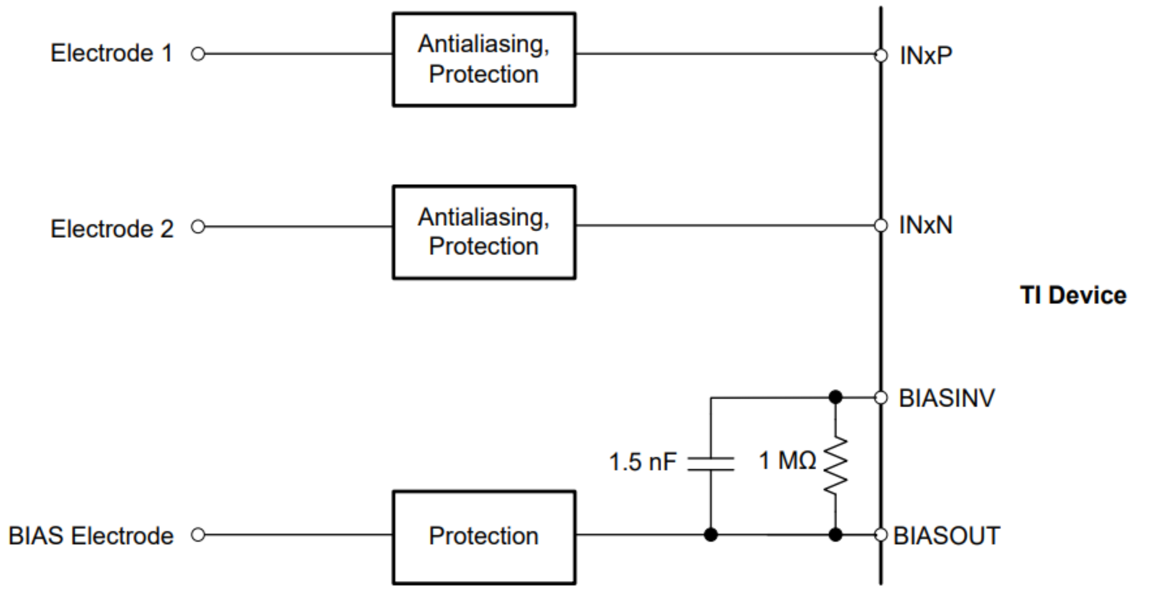
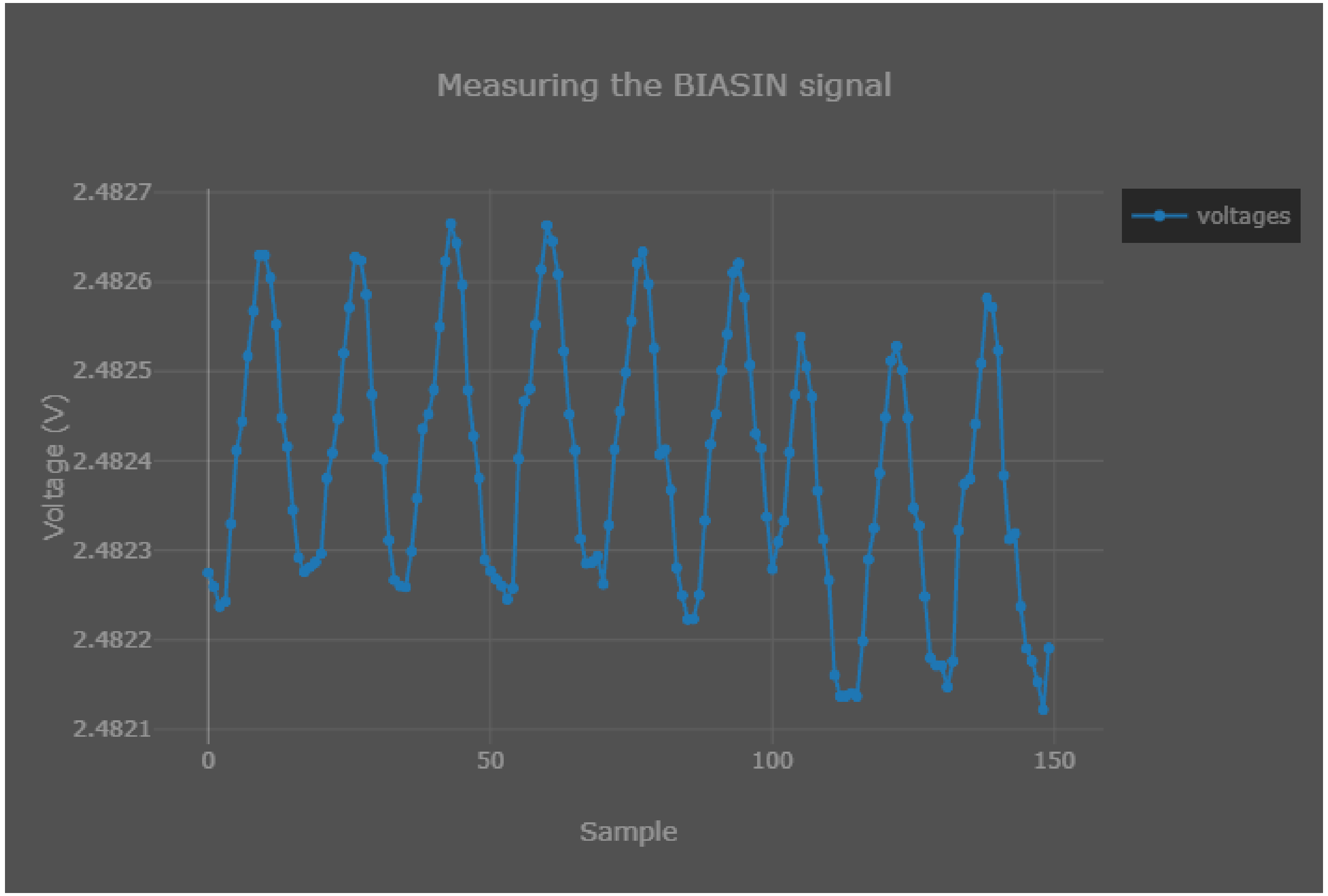
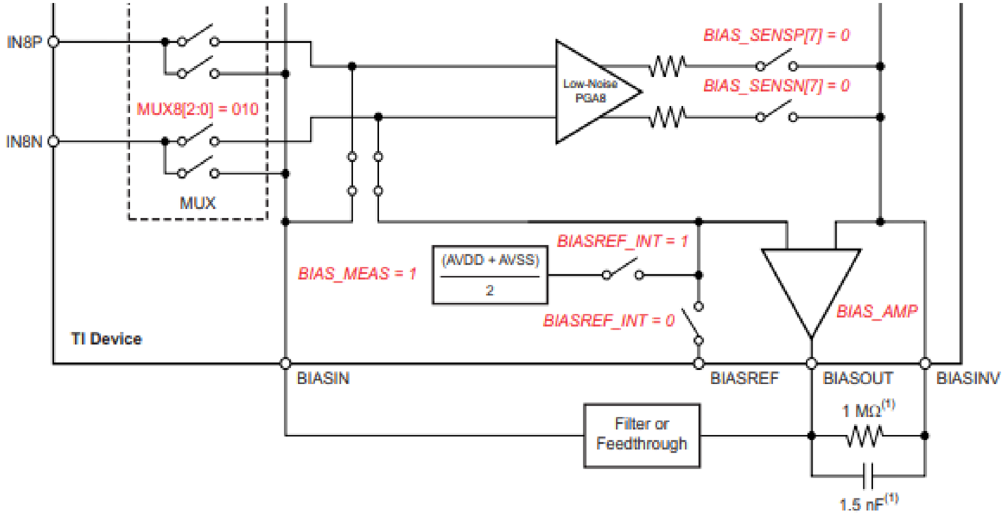


Figure 68. Setting Common-Mode Using BIAS Electrode

Observation / Conclusion



Right now, BIASIN and BIASOUT are shorted.
There is an option to route the BIASIN signal to a channel using MUX for measurement.



(1) Typical values for example only.

Figure 34. BIASOUT Signal Configured to be Read Back by Channel 8

Next steps

- Test the bias amplifier more
- Begin revising the schematic of the AFE circuit for the new design